

Smart Room Entry System

Project Aim

To automate room access, lighting, and occupancy-based fan control..

Project Overview

The Smart Room Entry System is an Arduino-based automation project that uses two Arduino Uno boards communicating through serial communication. An ultrasonic sensor detects visitors, automatically opens and closes a door using a servo motor, controls room lighting, monitors occupancy, and activates a fan when a visitor remains present for an extended period.

Components Used

1. Arduino Uno (Master) – 1
2. Arduino Uno (Slave) – 1
3. HC-SR04 Ultrasonic Sensor – 1
4. 16×2 LCD Display – 1
5. 10kΩ Potentiometer – 1
6. Servo Motor (SG90) – 1
7. DC Motor (Fan) – 1
8. PNP Transistor – 1
9. Red LED (Door Status Indicator) – 1
10. Green LED (Room Light Indicator) – 1
11. 220Ω Resistor – 2
12. 1kΩ Resistor – 1
13. Breadboard – 2
14. Jumper Wires – As RequiredResistors (220Ω)

Construction and Wiring Details

Component	Component Pin	Connected To
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Component	Component Pin	Connected To
HC-SR04 Ultrasonic Sensor	VCC	Arduino 1 - 5V
HC-SR04 Ultrasonic Sensor	GND	Arduino 1 - GND
HC-SR04 Ultrasonic Sensor	TRIG	Arduino 1 - D2
HC-SR04 Ultrasonic Sensor	ECHO	Arduino 1 - D3
LCD Display	RS	Arduino 1 - D4
LCD Display	E	Arduino 1 - D5
LCD Display	DB4	Arduino 1 - D6
LCD Display	DB5	Arduino 1 - D7
LCD Display	DB6	Arduino 1 - D8
LCD Display	DB7	Arduino 1 - D9
LCD Display	RW	GND
LCD Display	VCC	Arduino 1 - 5V
LCD Display	GND	Arduino 1 - GND
LCD Display	V0	Potentiometer Middle Pin
Potentiometer	Left Pin	GND
Potentiometer	Right Pin	5V
Potentiometer	Middle Pin	LCD V0
Arduino 1	D1 (TX)	Arduino 2 - D0 (RX)
Arduino 1	GND	Arduino 2 - GND
Servo Motor	Signal	Arduino 2 - D3
Servo Motor	VCC	Arduino 2 - 5V
Servo Motor	GND	Arduino 2 - GND

Component	Component Pin	Connected To
Red LED (Door Status)	Anode (+)	Arduino 2 - D4 through 220Ω resistor
Red LED (Door Status)	Cathode (-)	GND
White LED (Room Light)	Anode (+)	Arduino 2 - D6 through 220Ω resistor
White LED (Room Light)	Cathode (-)	GND
PNP Transistor	Base	Arduino 2 - D5 through 1kΩ resistor
PNP Transistor	Emitter	5V
PNP Transistor	Collector	DC Motor Terminal 2
DC Motor (Fan)	Terminal 1	GND
DC Motor (Fan)	Terminal 2	PNP Transistor Collector
Arduino 2	D0 (RX)	Arduino 1 - D1 (TX)
Arduino 2	GND	Arduino 1 - GND

[illegible]

Component		Component Pin Connected To	
Arduino 1		LCD Display	
Arduino 1		LCD Display	
Arduino 1		LCD Display	
Arduino 1		LCD Display	
Arduino 1	Potentiometer	Left Pin	GND
Arduino 1	Potentiometer	Right Pin	5V
Arduino 1	Serial Communication	TX (D1)	Arduino 2 RX (D0)
Arduino 1	Serial Communication	GND	Arduino 2 GND
Arduino 2	Servo Motor	Signal	D3
Arduino 2	Servo Motor	VCC	5V
Arduino 2	Servo Motor	GND	GND
Arduino 2	Door Status LED	Anode (+)	D4 through 220 Ω resistor
Arduino 2	Door Status LED	Cathode (-)	GND
Arduino 2	Room Light LED	Anode (+)	D6 through 220 Ω resistor
Arduino 2	Room Light LED	Cathode (-)	GND
Arduino 2	PNP Transistor	Base	D5 through 1k Ω resistor
Arduino 2	PNP Transistor	Emitter	5V
Arduino 2	PNP Transistor	Collector	DC Motor Terminal 2
Arduino 2	DC Motor (Fan)	Terminal 1	GND

Code Structure / Program Flow

START

|



Initialize:

- LCD

- Ultrasonic Sensor

- Serial Communication

- Servo Motor

- LEDs

- Fan

|

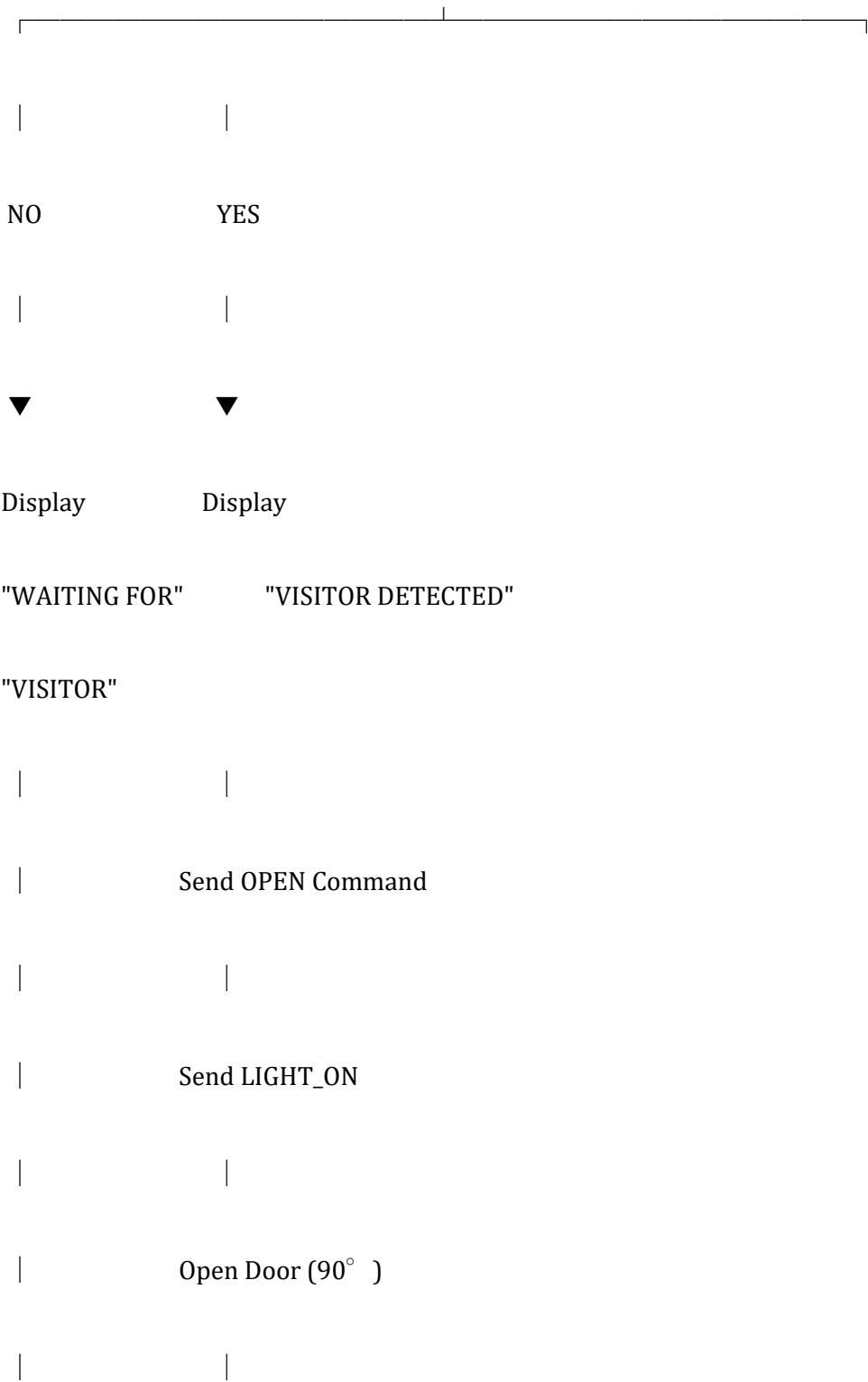


Measure Distance Using Ultrasonic Sensor

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Is Distance $\leq$ 15 cm?





NO YES



Continue Display

Monitoring "ROOM OCCUPIED"



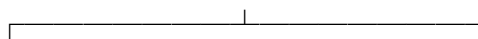
Send FAN_ON



Turn Fan ON



Visitor Leaves?



|

|

NO

YES

|

|



Keep Fan ON Send FAN_OFF

Send LIGHT_OFF

|



Turn Fan OFF

Turn Light OFF

|



Return to Monitoring

|



END

Source Code

Arduino 1

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(4, 5, 6, 7, 8, 9);
```

```
const int trigPin = 2;
```

```
const int echoPin = 3;
```

```
bool visitorDetected = false;
```

```
bool fanOn = false;
```

```
bool doorOpened = false;
```

```
unsigned long objectStartTime = 0;
```

```
void setup()
```

```
{
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
  Serial.begin(9600);
```

```
  lcd.begin(16, 2);
```

```
  lcd.print("SMART ROOM");
```

```
  delay(2000);
```

```
  lcd.clear();
```

```
}
```

```
void loop()
```

```
{
```

```
    long duration;
```

```
    float distance;
```

```
    digitalWrite(trigPin, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(trigPin, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(trigPin, LOW);
```

```
    duration = pulseIn(echoPin, HIGH);
```

```
    distance = duration * 0.034 / 2;
```

```
    if(distance <= 15)
```

```
    {
```

```
        // First detection
```

```
        if(!visitorDetected)
```

```
        {
```

```
            visitorDetected = true;
```

```
            objectStartTime = millis();
```

```
            lcd.clear();
```

```
    lcd.print("VISITOR");  
    lcd.setCursor(0,1);  
    lcd.print("DETECTED");  
  
    Serial.println("LIGHT_ON");  
  
    // Open door only once  
    if(!doorOpened)  
    {  
        Serial.println("OPEN");  
        doorOpened = true;  
    }  
}  
  
// Fan after 5 seconds  
if((millis() - objectStartTime >= 5000) && !fanOn)  
{  
    fanOn = true;  
  
    lcd.clear();  
    lcd.print("ROOM");  
    lcd.setCursor(0,1);  
    lcd.print("OCCUPIED");  
  
    Serial.println("FAN_ON");  
}  
}  
else
```

```
{  
  if(visitorDetected)  
  {  
    Serial.println("FAN_OFF");  
    Serial.println("LIGHT_OFF");  
  }
```

```
  visitorDetected = false;  
  fanOn = false;  
  doorOpened = false;  
  objectStartTime = 0;
```

```
  lcd.clear();  
  lcd.print("WAITING FOR");  
  lcd.setCursor(0,1);  
  lcd.print("VISITOR");  
}
```

```
  delay(100);  
}
```

Arduino 2

```
#include <Servo.h>
```

```
Servo door;
```

```
const int servoPin = 3;  
const int doorLed = 4;
```

```
const int fanPin = 5;
```

```
const int roomLight = 6;
```

```
String command = "";
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
  pinMode(doorLed, OUTPUT);
```

```
  pinMode(fanPin, OUTPUT);
```

```
  pinMode(roomLight, OUTPUT);
```

```
  digitalWrite(doorLed, LOW);
```

```
  digitalWrite(fanPin, LOW);
```

```
  digitalWrite(roomLight, LOW);
```

```
  door.attach(servoPin);
```

```
  door.write(0);
```

```
}
```

```
void loop()
```

```
{
```

```
  if(Serial.available())
```

```
  {
```

```
    command = Serial.readStringUntil('\n');
```

```
    command.trim();
```

```
if(command == "OPEN")
{
    digitalWrite(doorLed, HIGH);

    door.write(90);

    delay(3000);

    door.write(0);

    digitalWrite(doorLed, LOW);
}

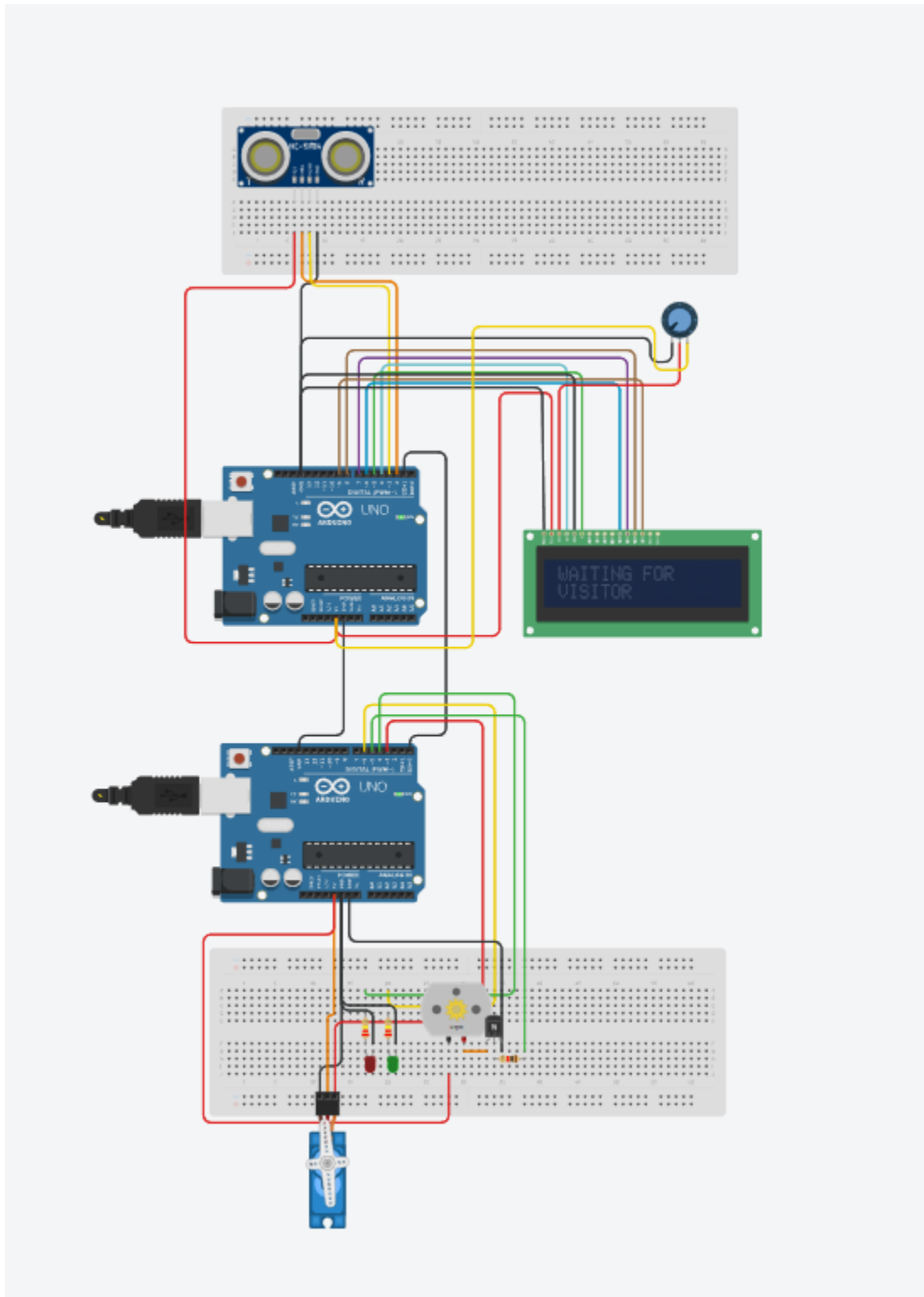
else if(command == "LIGHT_ON")
{
    digitalWrite(roomLight, HIGH);
}

else if(command == "LIGHT_OFF")
{
    digitalWrite(roomLight, LOW);
}

else if(command == "FAN_ON")
{
    digitalWrite(fanPin, HIGH);
}
```

```
else if(command == "FAN_OFF")  
{  
    digitalWrite(fanPin, LOW);  
}  
}
```

Project Photograph



Links

Google drive :

https://drive.google.com/drive/folders/1RwxHWS914YHTI3d6lFceaCnzw1gLWrdc?usp=drive_link

TinkerCad :

<https://www.tinkercad.com/things/hIzGfgMfeRO/editel?returnTo=%2Fdashboard&sharecode=WijKw2v6CcvI9w-ll0kAsg9TtwuQPQpqrqxH2oM8eRg>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>